

# IPW3 : NACA0012 Code-to-Code Comparisons

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## Geometry and Aerodynamic Conditions



### NASA Icing Research Tunnel

|                 |           |
|-----------------|-----------|
| Airfoil         | NACA0012  |
| Chord, $c$      | 0.5334 m  |
| Angle of attack | $4^\circ$ |

### Aerodynamic Conditions

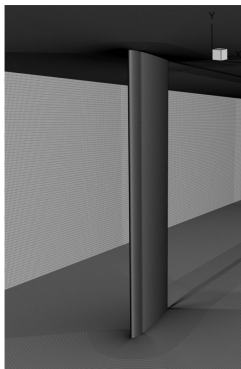
|               |                          |
|---------------|--------------------------|
| $M_\infty$    | 0.3114                   |
| $U_\infty$    | 102.15 m/s               |
| $T_\infty$    | 267.75 K                 |
| $\rho_\infty$ | 1.1884 kg/m <sup>3</sup> |
| $Re_c$        | $3.8441 \times 10^6$     |

### Quantities of Interest

$$C_L, \quad C_D, \quad C_M, \quad C_p$$



## Icing Conditions



| Grid | Nodes   | Elements |
|------|---------|----------|
| L1   | 60.45 M | 59.86 M  |
| L2   | 27.31 M | 26.99 M  |
| L3   | 11.23 M | 11.08 M  |
| L4   | 5.95 M  | 5.86 M   |

|                          | AE3932       | AE3933      |
|--------------------------|--------------|-------------|
| $T_0$ [°C]               | -5.6         | -2.2        |
| $T_\infty$ [K]           | 262.305      | 265.8       |
| $M_\infty$               | 0.3151       | 0.3127      |
| $U_\infty$ [m/s]         | 102.31       | 102.19      |
| LWC [g/m <sup>3</sup> ]  | 0.55         | 0.55        |
| MVD [ $\mu$ m]           | 20           | 20          |
| Exposure time [min]      | 7            | 7           |
| <b>Exp. ice mass [g]</b> | <b>101.0</b> | <b>85.9</b> |

### Experimental Observation

The warmer AE3933 condition produces approximately **15% less ice mass** than AE3932 (85.9 g vs. 101.0 g). This difference will be revisited when assessing the predicted thermodynamic mass balance and ice accretion.

### Droplet Distribution

- Single-bin to 15-bin distributions
- Total LWC preserved between distributions
- Evaluated across all four grid levels

### Quantities of Interest

$$\beta, \quad HTC, \quad T_s, \quad f, \quad m_{\text{water}}, \quad m_{\text{ice}}, \quad m_{\text{evap}}$$

## NACA0012 – Modeling Choices Across Submissions

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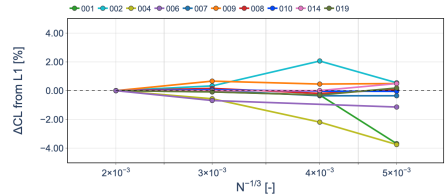
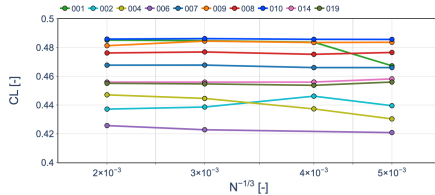


## NACA0012 – Summary of Data Received

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## NACA0012 – Lift Coefficient Grid Convergence

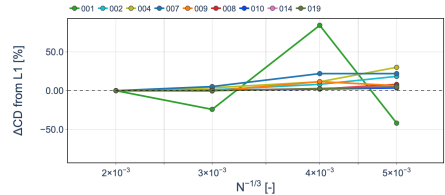
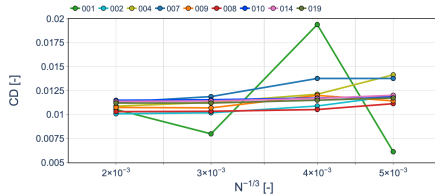


### Assessment

- At L1, the median lift coefficient is  $C_L = 0.4677$ , with an IQR of 0.0262 (5.6%).
- Most submissions exhibit limited sensitivity to spatial grid refinement.
- Relative to L1, the largest coarse-grid deviations are approximately 4%.



## NACA0012 – Drag Coefficient Grid Convergence

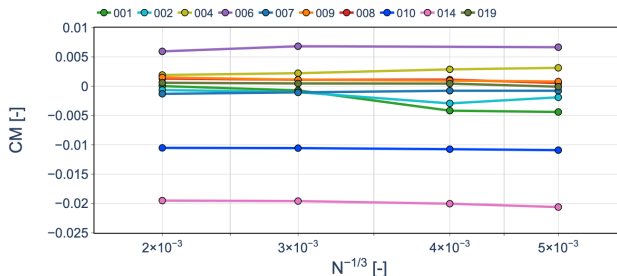


### Assessment

- At L1, the median drag coefficient is  $C_D = 0.01088$ , with an IQR of  $7.57 \times 10^{-4}$  (7.0%).
- $C_D$  exhibits greater sensitivity to spatial grid refinement than  $C_L$ .
- Non-monotonic grid-convergence behavior is observed for selected submissions.



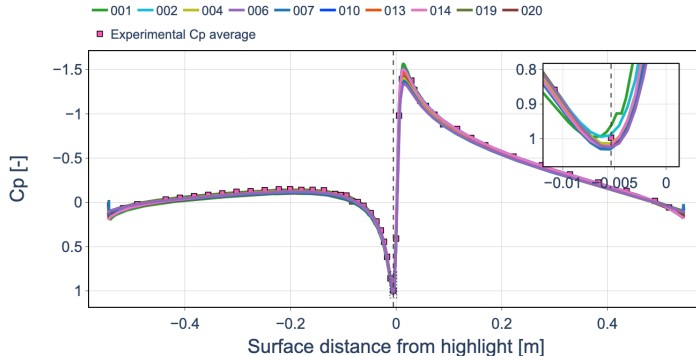
## NACA0012 – Pitching Moment Grid Convergence



### Assessment

- At L1, the median pitching-moment coefficient is  $C_M \approx 0$ , with an IQR of  $2.60 \times 10^{-3}$ .
- Most individual submissions exhibit limited sensitivity to spatial grid refinement.
- Code-to-code offsets are larger than the grid-refinement effect for most submissions.
- Relative percentage differences are not meaningful for  $C_M$  because the reference values are close to zero.

## NACA0012 – Surface Pressure Distribution (L1)



## Assessment

- The submitted  $C_p$  distributions are closely clustered and show good agreement with the experimental measurements over most of the airfoil.
- The largest code-to-code differences are concentrated near the leading-edge suction peak and stagnation region.
- The overall pressure distribution is consistent across submissions despite differences in prescribed surface roughness.

## NACA0012 – Aerodynamic Assessment Summary

### Grid Convergence

- $C_L$ : generally weak grid sensitivity, with most submissions remaining close to their L1 prediction.
- $C_D$ : greater grid sensitivity, with non-monotonic behavior observed for selected submissions.
- $C_M$ : generally limited within-code grid sensitivity, while code-to-code offsets remain comparatively large.

### Code-to-Code Variability at L1

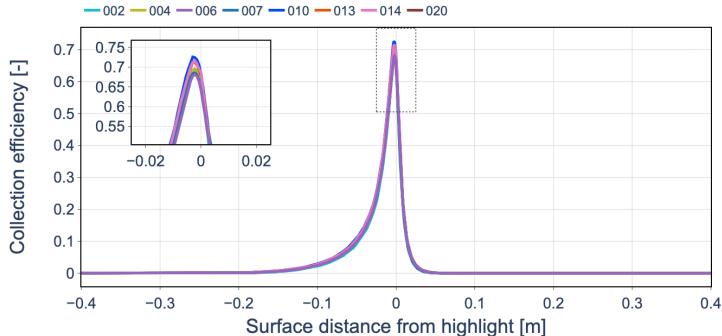
|       | Median      | IQR                   | Relative IQR |
|-------|-------------|-----------------------|--------------|
| $C_L$ | 0.4677      | 0.0262                | 5.6%         |
| $C_D$ | 0.01088     | $7.57 \times 10^{-4}$ | 7.0%         |
| $C_M$ | $\approx 0$ | $2.60 \times 10^{-3}$ | –            |

*Relative IQR is not reported for  $C_M$  because normalization by a value close to zero is ill-conditioned.*

### Aerodynamic Takeaway

The  $C_p$  distributions are closely clustered and generally agree well with the experimental measurements. Grid refinement produces relatively small changes in  $C_L$  and  $C_M$  for most submissions, while  $C_D$  is more grid-sensitive. Residual code-to-code differences therefore cannot be attributed to spatial resolution alone.

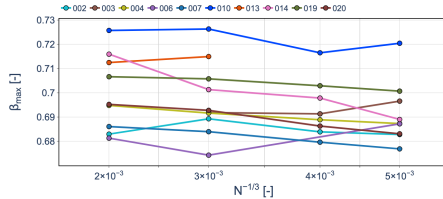
## NACA0012 – Collection Efficiency



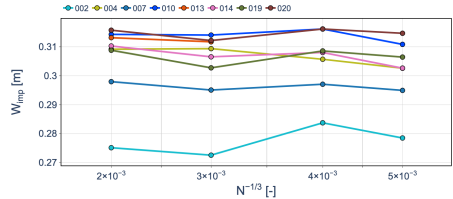
## Assessment

- Most submissions predict a similar collection-efficiency distribution around the leading edge.
- Peak collection efficiencies are closely clustered, while differences remain in the detailed profile and impingement extent.
- Differences in prescribed surface roughness should be considered when interpreting the remaining code-to-code variability.

## NACA0012 – Collection Efficiency Grid Convergence



Peak Collection Efficiency



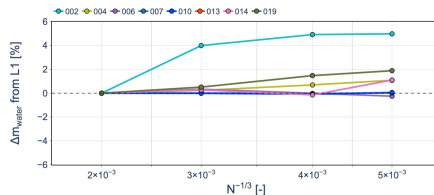
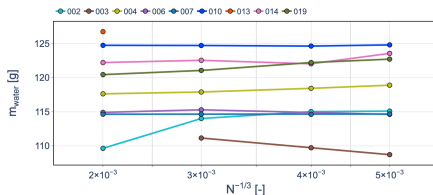
Impingement Width

### Assessment

- At L1, the median peak collection efficiency is  $\beta_{\max} = 0.695$  with an IQR of 0.0209 (3.0%).
- The L1 median impingement width is 0.309 m with an IQR of  $8.93 \times 10^{-3}$  m (2.9%).
- Both quantities exhibit limited grid sensitivity for most submissions, indicating comparatively stable local impingement predictions.



## NACA0012 – Water Mass Grid Convergence

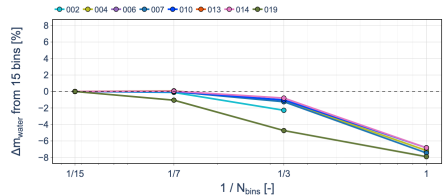
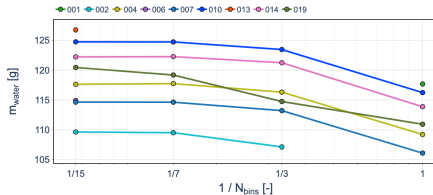


### Assessment

- For the 15-bin distribution, the L1 median water mass is 119.05 g with an IQR of 6.39 g (5.37%).
- Most submissions exhibit weak grid sensitivity, with changes generally within approximately 5% of their L1 prediction.
- Code-to-code differences are generally larger than the grid-refinement effect.



## NACA0012 – Droplet Distribution Sensitivity



## Assessment

- The 15-bin and 7-bin distributions produce very similar impinged water masses.
- Reducing the distribution from 7 to 3 bins introduces only a modest change for most submissions.
- Relative to the 15-bin distribution, the single-bin approximation reduces the collected water mass by approximately 5–8% for most participants.



## NACA0012 – Droplet Impingement Summary

### Spatial Grid Convergence

- Limited grid sensitivity is observed for most submissions.
- Code-to-code variability generally exceeds the grid-refinement effect.

### Droplet-Size Distribution

- The 7- and 15-bin distributions provide very similar results.
- The single-bin approximation predicts approximately 5–8% less collected water.

### Icing-Condition Sensitivity

- Little difference in total water impingement is observed between the two icing conditions.

### Takeaway

Grid refinement has a limited effect, while droplet-size discretization introduces a more systematic sensitivity. A 7-bin distribution closely reproduces the 15-bin result.



## NACA0012 – Convective Heat Transfer Evaluation

### Integrated Convective Heat Transfer

The convective heat transfer is evaluated along the surface slice as

$$Q'_c = \int HTC(s) [T_s(s) - T_{rec}(s)] ds \quad [Q'_c] = \text{W/m}.$$

The submitted  $HTC$ , surface temperature  $T_s$ , and pressure coefficient  $C_p$  are used directly.

### Recovery Temperature Reconstruction

Since  $T_{rec}$  is not directly available from the submissions, it is reconstructed from the submitted  $C_p$  distribution:

$$C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{rec}(s)$$

Assuming  $p_s \simeq p_e$  and an isentropic external flow,

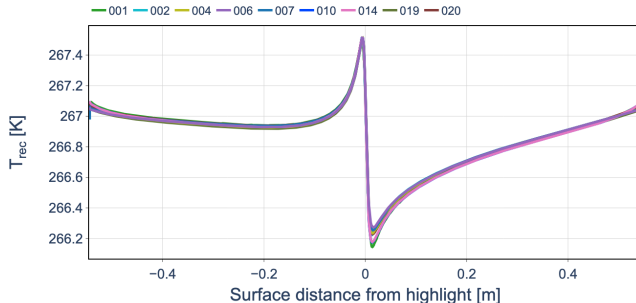
$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_{p,e}.$$

### Recovery Model

For a turbulent boundary layer,

$$T_{rec} = T_e \left[ 1 + r \frac{\gamma - 1}{2} M_e^2 \right], \quad r = Pr^{1/3} \approx 0.9.$$

## NACA0012 – Recovery Temperature Distribution (L1)

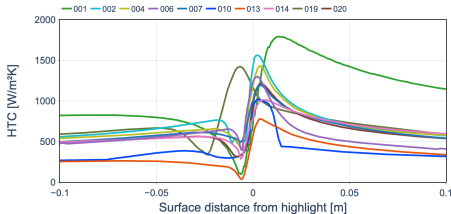


## Assessment

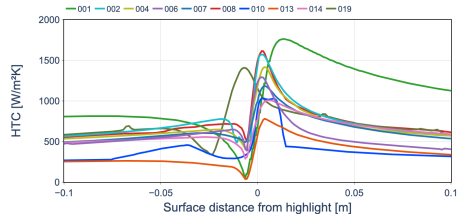
- Reconstructed recovery-temperature distributions are closely clustered across submissions.
- This good agreement is consistent with the close agreement previously observed in the  $C_p$  distributions.



## NACA0012 – Convective Heat-Transfer Coefficient (L1)



**AE3932**



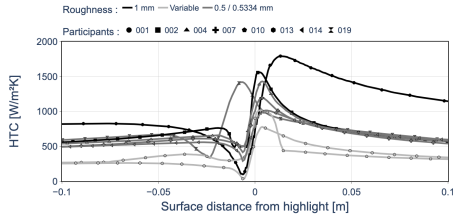
**AE3933**

### Assessment

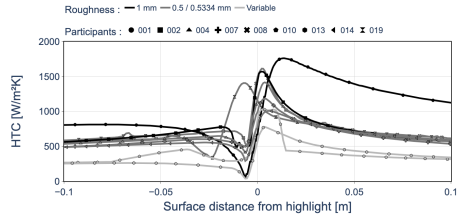
- Considerable code-to-code variability is observed in  $HTC$ , particularly near the leading edge, with similar behavior for both conditions.
- Grouping by prescribed roughness shows that roughness contributes to the observed variability, but does not fully explain the code-to-code differences.



## NACA0012 – Convective Heat-Transfer Coefficient (L1)



**AE3932 – Grouped by Roughness**



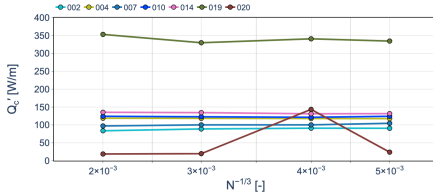
**AE3933 – Grouped by Roughness**

### Assessment

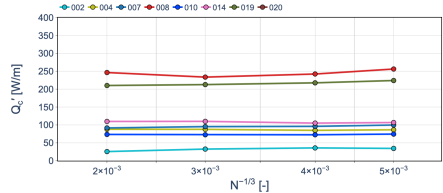
- Considerable code-to-code variability is observed in  $HTC$ , particularly near the leading edge, with similar behavior for both conditions.
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## NACA0012 – Convective Heat Transfer Grid Convergence



**AE3932**



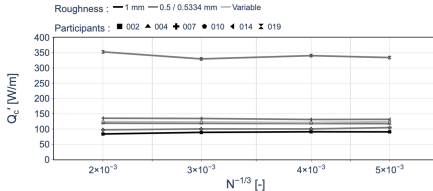
**AE3933**

### Assessment

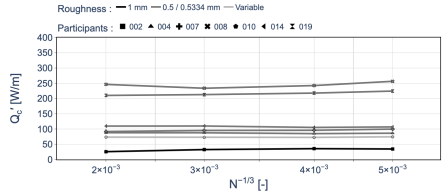
- Most submissions exhibit limited grid sensitivity, while code-to-code variability is substantially larger than the grid-refinement effect.
- AE3932 and AE3933 show broadly comparable integrated convective heat-transfer predictions.
- Grouping by prescribed roughness shows a systematic roughness effect, although substantial code-to-code variability remains.



# NACA0012 – Convective Heat Transfer Grid Convergence



**AE3932 – Grouped by Roughness**



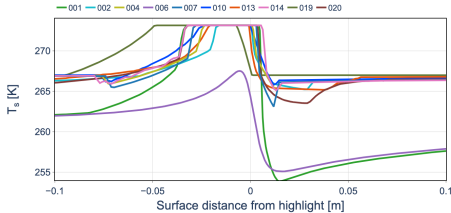
**AE3933 – Grouped by Roughness**

## Assessment

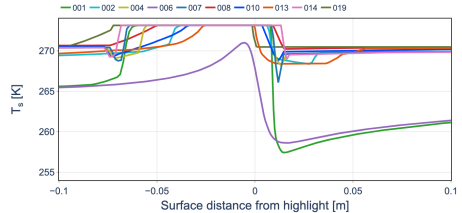
- Most submissions exhibit limited grid sensitivity, while code-to-code variability is substantially larger than the grid-refinement effect.
- AE3932 and AE3933 show broadly comparable integrated convective heat-transfer predictions.
- Grouping by prescribed roughness shows a systematic roughness effect, although substantial code-to-code variability remains.



## NACA0012 – Surface Temperature Comparison



**AE3932**



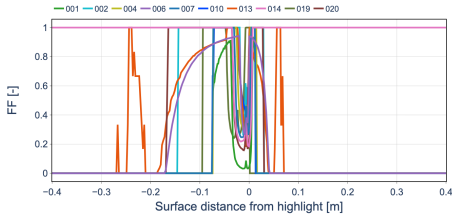
**AE3933**

### Assessment

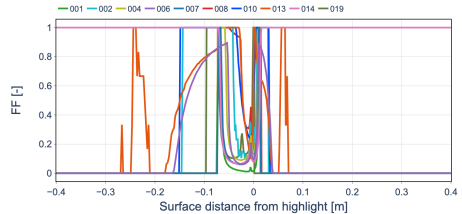
- Substantial code-to-code variability is observed in the surface-temperature distributions for both conditions.
- Differences between AE3932 and AE3933 are also evident in the predicted surface thermal response.



## NACA0012 – Freezing Fraction Comparison



**AE3932**



**AE3933**

### Assessment

- Considerable code-to-code variability is observed in the freezing fraction for both conditions.
- Different thermodynamic regimes and transition locations are predicted across submissions.
- Sharp and spatially shifted transitions make direct pointwise comparison particularly difficult.



## NACA0012 – HTC & Thermodynamics Assessment

### Recovery Temperature

- Closely clustered across submissions, consistent with the good agreement in  $C_p$ .

### Convective Heat Transfer

- Larger code-to-code variability is observed in  $HTC$  and  $Q'_c$ , with roughness explaining only part of the spread.
- Grid sensitivity remains comparatively limited for most submissions.

### Thermodynamic Response

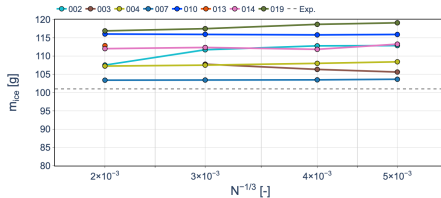
- Surface temperature and freezing fraction exhibit substantial code-to-code variability.
- Different freezing regimes and transition locations are predicted across submissions.

### Takeaway

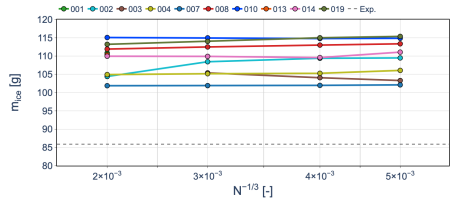
Aerodynamic agreement leads to consistent  $T_{rec}$ , while larger differences emerge in  $HTC$  and propagate into the predicted thermodynamic response.



## NACA0012 – Ice Mass Grid Convergence



**AE3932**



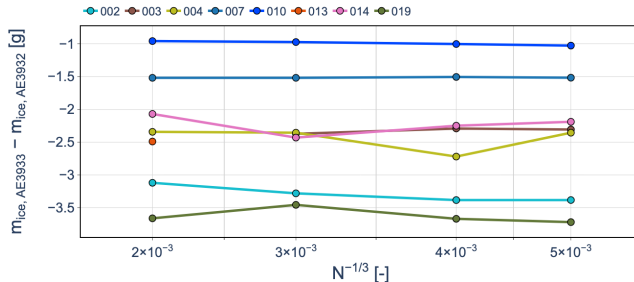
**AE3933**

### Assessment

- Most submissions exhibit limited ice-mass sensitivity to spatial grid refinement for the 15-bin distribution.
- An isolated submission exhibits substantially stronger grid sensitivity than the main participant cluster.
- Code-to-code differences remain visible on the finest grid and cannot be attributed solely to spatial resolution.
- A systematic difference between AE3932 and AE3933 begins to emerge in the predicted ice mass.



## NACA0012 – Ice Mass Difference Between Conditions



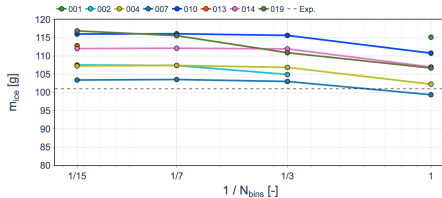
## Assessment

- The matched-participant comparison shows lower ice mass for AE3933 than AE3932 for most submissions.
- The difference remains comparatively consistent across the grid hierarchy for most submissions.

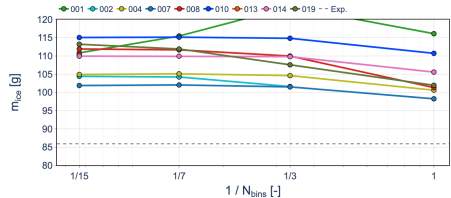
## Key Observation

A systematic difference between the two conditions emerges in ice mass; however, the numerical predictions do not reproduce the approximately 15% difference observed experimentally.

## NACA0012 – Ice Mass Sensitivity to Droplet Distribution



**AE3932**

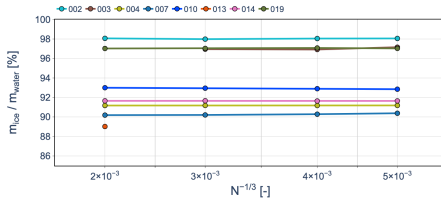


**AE3933**

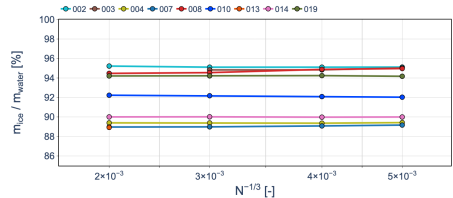
### Assessment

- The 15- and 7-bin distributions generally produce similar ice masses.
- Greater sensitivity appears as the droplet distribution is reduced to 3 and particularly 1 bin.
- For most submissions, reducing the droplet distribution modifies the ice mass more noticeably than the 15-to-7-bin reduction.
- The response is participant-dependent, indicating that droplet-size discretization influences the resulting icing response in addition to total water impingement.

## NACA0012 – Ice-to-Water Mass Ratio



**AE3932**



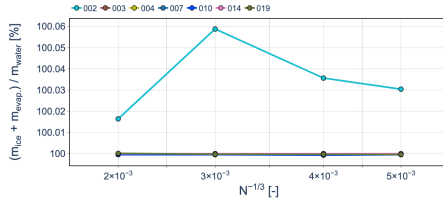
**AE3933**

### Assessment

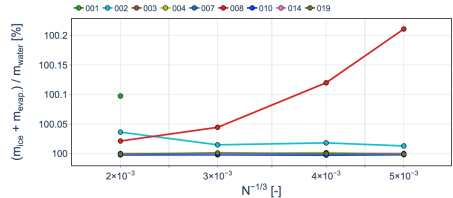
- Normalizing by the impinged water mass separates the icing response from differences in total water collection.
- Since the two conditions exhibit comparable water impingement, differences in this ratio directly highlight differences in the resulting ice accretion.
- The comparison therefore provides a clearer measure of the transition from impingement to the thermodynamic icing response.



## NACA0012 – Ice and Evaporation-to-Water Mass Ratio



AE3932



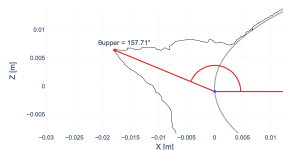
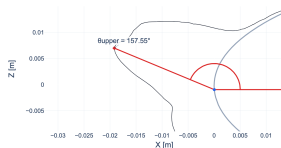
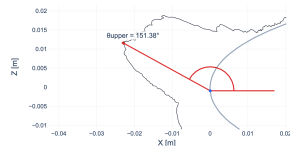
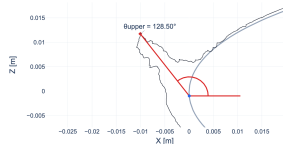
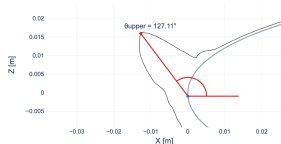
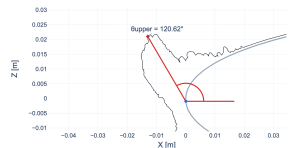
AE3933

## Assessment

- Including evaporated water provides additional information on the partition of the impinged water mass.
- Comparison with  $m_{ice}/m_{water}$  identifies the contribution of evaporation to the difference between the two icing conditions.
- The remaining fraction represents water leaving the local ice-accretion balance through the other modeled mass-transfer mechanisms.

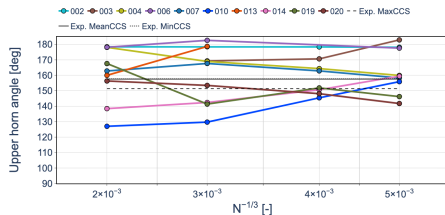


## NACA0012 – Experimental Upper Horn Angle

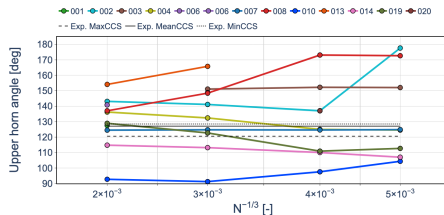
**MinCCS****MeanCCS  
AE3932****MaxCCS****MinCCS****MeanCCS  
AE3933****MaxCCS****Assessment**

- The experimental upper-horn angle is approximately  $151\text{--}158^\circ$  for AE3932 and  $121\text{--}129^\circ$  for AE3933.
- The experiments therefore indicate a clear change in upper-horn orientation between the two conditions.

## NACA0012 – Upper Horn Angle Grid Convergence



AE3932



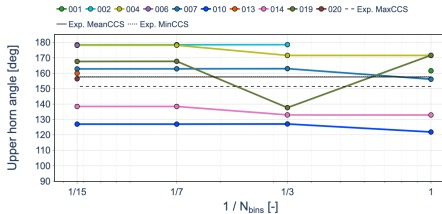
AE3933

## Assessment

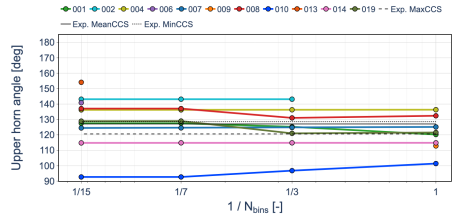
- The upper horn angle provides a geometric grid-convergence metric complementary to the integrated ice mass.
- Differences in horn orientation quantify changes in where the accreted ice is distributed rather than only how much ice is formed.
- Comparison between AE3932 and AE3933 indicates whether the change in icing response also produces a systematic change in horn orientation.



# NACA0012 – Upper Horn Angle Sensitivity to Droplet Distribution



AE3932



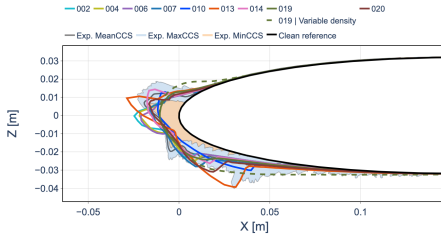
AE3933

## Assessment

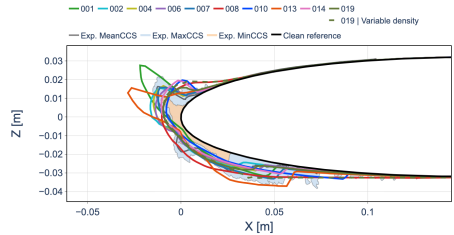
- The L1 comparison evaluates the sensitivity of the upper horn orientation to droplet-size discretization.
- This provides a geometric counterpart to the ice-mass droplet-distribution sensitivity.
- Changes in total ice mass and changes in horn orientation should therefore be considered together when assessing droplet-distribution convergence.



## NACA0012 – Ice Shapes (L1, 15 Bins)



**AE3932**



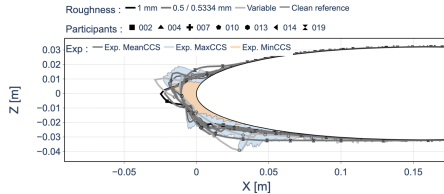
**AE3933**

### Assessment

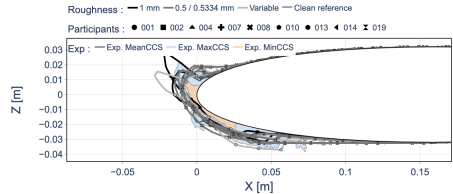
- Considerable code-to-code variability is observed in the predicted ice shapes for both conditions.
- Experimental ice shapes provide the validation reference for the predicted accretion geometry.
- Grouping by prescribed roughness helps assess the contribution of roughness to the observed geometric variability.



## NACA0012 – Ice Shapes (L1, 15 Bins)



**AE3932 – Grouped by Roughness**



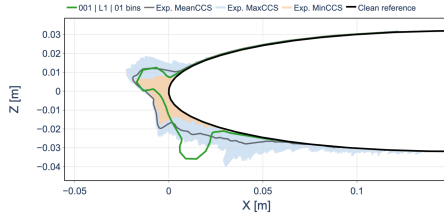
**AE3933 – Grouped by Roughness**

### Assessment

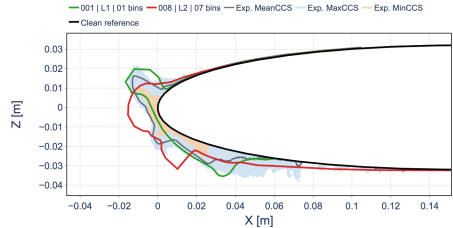
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## NACA0012 – Multilayer Ice-Shape Comparison (L1)



**AE3932**



**AE3933**

### Assessment

- The multilayer predictions are compared directly with the experimental ice-shape envelopes for both conditions.
- Differences are observed in the predicted horn extent, orientation, and local ice thickness.
- The multilayer comparison provides the final assessment of the evolving ice-accretion methodology.



## NACA0012 – Ice Accretion Summary

### Ice Mass

- Most submissions show limited grid sensitivity, while a systematic difference emerges between AE3932 and AE3933.

### Mass Partition

- Mass ratios highlight differences in freezing and evaporation despite comparable water impingement.

### Ice Geometry

- Horn metrics and ice shapes characterize how differences in ice mass translate into the final accretion geometry.

### Takeaway

While aerodynamics, impingement, and heat transfer remain broadly comparable, the resulting ice accretion distinguishes AE3932 and AE3933.



## NACA0012 – Overall Assessment

### Aerodynamics

- Limited grid sensitivity overall, with larger sensitivity in  $C_D$ ; L1  $C_p$  agrees well with experiment.

### Droplet Impingement

- Limited grid sensitivity and similar impingement between conditions; droplet-size discretization has a more systematic effect.

### Convective Heat Transfer

- Recovery temperature is closely clustered, while larger code-to-code variability is observed in  $HTC$ .

### Ice Accretion

- The clearest distinction between AE3932 and AE3933 emerges in the resulting ice accretion.

